WORKED SOLUTION

STEP III 2008, Question 11

Mechanics • Rotational Dynamics

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Read the question before you go on. It is in the STEP Support Programme's question database:

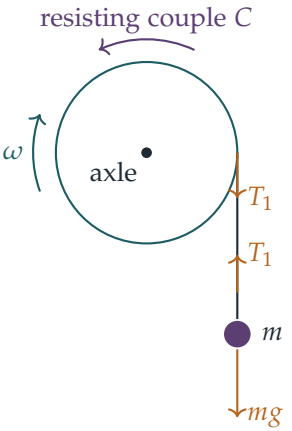
step.maths.org/questions/08-s3-q11

The key idea

A constant couple means constant angular deceleration, so once the string has gone the wheel obeys the rotational version of the "suvat" equations, and the first result is just "distance = average speed × time". That same stage tells us the size of the resisting couple in terms of I. For the descent, an energy equation is the natural tool: we know how far everything moves, and work done against a constant couple is simply couple × angle.

Solution

Model. The wheel is a rigid body turning about a fixed smooth axle, except for a resisting couple of constant magnitude C, which acts throughout the motion (while the particle descends as well as afterwards). The string is light and inextensible and does not slip, so while it is attached the particle's speed is $v = r\omega$, where ω is the wheel's angular speed.



The string leaves the right-hand side of the wheel, so the wheel turns clockwise as the particle falls, and the couple C acts anticlockwise, opposing the motion. (We won't actually need the tension T_1 in the string, because we use energy.)

After the string detaches

Now the only couple on the wheel is C, so taking moments about the axle,

$$I\dot{\omega} = -C,$$

and the angular deceleration C/I is constant. The angular speed therefore falls *linearly* from ω_0 to 0 in time T, so

$$\omega_0 = \frac{C}{I}\,T, \qquad \text{that is,} \qquad C = \frac{I\omega_0}{T}. \tag{1}$$

With a linear fall in angular speed, the average angular speed is $\frac{1}{2}(\omega_0 + 0)$, and the angle turned is the average angular speed times the time. That angle is n_2 revolutions, or $2\pi n_2$ radians, so

$$\tfrac{1}{2}\omega_0 T = 2\pi n_2,$$

which is the first result. (It is the rotational version of $s = \frac{1}{2}(u+v)t$.) It also gives

$$\omega_0 = \frac{4\pi n_2}{T}, \qquad \text{and so by (1)} \qquad C = \frac{4\pi n_2 I}{T^2}. \tag{2}$$

While the particle descends

While the wheel turns through n_1 revolutions, that is $2\pi n_1$ radians, the string unwinds a length $2\pi n_1 r$, so the particle falls a distance $2\pi n_1 r$. At the moment of detachment the wheel has angular speed ω_0 and, as the string has not slipped, the particle has speed $r\omega_0$.

? **Ask yourself** Why can we use energy here, when there is a resistance?

Energy is not conserved, but the work–energy principle still holds: the loss of potential energy equals the gain in kinetic energy plus the work done against resistance. The work done against a constant couple C while turning through angle θ is $C\theta$, just as the work against a constant force is force $\times$ distance. The tension does no net work on the system, since the string is light and inextensible and does not slip, so the work it does on the wheel is exactly the work it takes from the particle.

So, from release to detachment,

$$mg\,(2\pi n_1 r) = \tfrac{1}{2}m(r\omega_0)^2 + \tfrac{1}{2}I\omega_0^2 + C\,(2\pi n_1).$$

Now substitute ω_0 and C from (2):

$$2\pi mgrn_1 = \tfrac{1}{2}\,(mr^2 + I)\,\frac{16\pi^2 n_2^2}{T^2} + 2\pi n_1 \cdot \frac{4\pi n_2 I}{T^2}.$$

Multiplying through by $T^2/(2\pi)$,

$$\begin{aligned} mgrn_1T^2 &= 4\pi n_2^2\,(mr^2 + I) + 4\pi n_1 n_2 I \\ &= 4\pi mr^2 n_2^2 + 4\pi n_2(n_1 + n_2)\,I. \end{aligned}$$

Rearranging, and dividing by $4\pi n_2(n_1 + n_2)$, which is positive,

$$I = \frac{mgrn_1T^2 - 4\pi mr^2 n_2^2}{4\pi n_2(n_1 + n_2)},$$

as required.

Checks

- *Dimensions.* Both terms in the numerator, $mgrT^2$ and mr^2, have the units of a moment of inertia ($\mathrm{kg\,m^2}$), and n_1, n_2 are pure numbers.
- *Numerical check.* We stepped the motion with $I = 0.5\,\mathrm{kg\,m^2}$, $m = 1\,\mathrm{kg}$, $r = 0.2\,\mathrm{m}$, $C = 0.5\,\mathrm{N\,m}$, $g = 9.8\,\mathrm{m\,s^{-2}}$ and $n_1 = 5$. This gives $\omega_0 \approx 13.03\,\mathrm{rad\,s^{-1}}$, $T \approx 13.03\,\mathrm{s}$ and $n_2 \approx 13.52$. Then $\frac{1}{2}\omega_0 T \approx 84.94 \approx 2\pi n_2$, and the formula returns $I = 0.5000$.
- *No resistance.* If $C \to 0$ the wheel never stops, so T and n_2 both become infinite; the formula behaves sensibly, since its numerator grows like T^2 and its denominator like $n_2^2 \propto T^2$.

Common slips

- Leaving out the work done against resistance during the descent. The question says the couple is constant, and it acts all the time, not just after the string detaches. Without it the formula comes out wrong: in the example above, it would give $I \approx 0.39$ instead of 0.5.
- Forgetting the particle's kinetic energy $\frac{1}{2}mr^2\omega_0^2$; this is where the $-4\pi mr^2 n_2^2$ term comes from.
- Mixing revolutions and radians: n revolutions is $2\pi n$ radians.

Beyond STEP

This is a neat experimental method: by timing the run-down of a wheel and counting revolutions, you can measure its moment of inertia *and* the friction in its bearing at the same time, even though you don't know either in advance. Undergraduate mechanics builds this into the full theory of rigid-body rotation, where angular momentum and torque play the roles of momentum and force, and the moment of inertia becomes a tensor for bodies spinning about arbitrary axes. Kleppner and Kolenkow, *An Introduction to Mechanics*, develops this carefully, with many problems of exactly this flavour.

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